(for Medium to large project) - JS good for small projects.

TYPESCRIPT → build on top of JS used with both frontend/backend

↳ JS + Types (type safety)

⇒ JS ⇒	⇒ TS ⇒
function add(a, b) { return a+b; }	function add(a : number, b : number) : number { return a+b; } (Typed function; typed parameter; typed return)
add(5, "10");	add(5, "10");
output -: 510	
JS allow this mistake	TS show error befor running
Problem with JS ↳ • No types safety • Bugs found at runtime • Hard to manage large project	Typescript solve this ↳ • Adding types • Better IDE support • Better code maintainability • Refactoring / new feature / code completion It prevents wrong assignments

we know type of variable at compile time

☆ Typescript is a statically typed superset of Javascript that compiles to Javascript and helps catch error during development.

-: Browser do not run TS directly

TS → Compiler → JS → Browser

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⇒ TS - lets you define what type of data a variable can store

string - let name : string = "Mudit";

number - let age : number = 22;

boolean - let isLoggedIn : boolean = true;

⇒ any (avoid using can store anything) - let data : any = "Hello"; - no type safety
data : 10;
data = true; // if not defined this is by default in TS

⇒ Unknown (Safer) - let value : unknown = "Hello"; - you must check type before using
if (typeof value === "string") {
console.log (value.toUpperCase());
} → type guard / type narrowing

void (used when fn returns nothing) - function logMessage() : void {
console.log ("Hello");
}

⇒ never (if fn never finish normally) - function throwError() : never {
throw new Error ("something went wrong");
}

Interfaces - Structure of an object.

interface User {
name : string;
age : number;
isAdmin : boolean;
}

use → const user : User = {
name : "Mudit",
age : 22,
isAdmin : false
};

if miss a property ts error.